How Does Revolution Affect Long-Term Economic Outcomes? Evidence from the Cultural Revolution in China*

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ABSTRACT

This study examines the effect of the Cultural Revolution on China's long-term economic outcomes – specifically, GDP output and agricultural productivity. The Cultural Revolution (1966 – 1976) was a chaotic period of social upheavals intended to preserve Chinese communism and reduce inequality by eliminating capitalism and traditional values in Chinese society. Utilizing datasets from Walder (2020) and the National Bureau of Statistics of China, I investigate the causal effects of revolution intensity on economic performance by using radio signals as an instrumental variable. Findings based on the IV regression models show that the Cultural Revolution had significant and enduring effects on both GDP output and agricultural productivity. These results suggest that potential policies can be applied to enhance regional development by identifying factors that contributed to economic recovery after the Revolution. However, the reliability of radio signals as a strong instrumental variable remains inconclusive, highlighting a need for future research to explore additional control variables to strengthen the causal inference.

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† Department of Economics, University of Wisconsin-Madison. [redacted]

1. Introduction

The Cultural Revolution (1966–1976) is a defining event in modern Chinese history. Driven by Chairman Mao Zedong's radical vision of enforcing communism, it sought to eliminate capitalist elements and remove entrenched traditional values. During this decade-long upheaval, millions of lives were disrupted, traditional structures were dismantled, and socio-political chaos harmed the country. While historians and political scientists have explored the socio-political consequence of this period, its long-term economic consequences are less understood. Understanding the economic legacy of the Cultural Revolution is crucial because it helps us comprehend the far-reaching impacts of political instability on economic structures. This study is not only timely but also pivotal for modern economic theory, as similar sociopolitical movements continue to affect economies globally. A thorough analysis can guide future policy in fostering economic resilience when facing political disruption.

Existing research on the economic impact of the Cultural Revolution is relatively sparse, with most scholarly attention focused on the sociopolitical dimensions. Walder (2020) laid a foundation by documenting the political, social, and economic upheavals during this period. He explored the localized impact of the revolution and collected valuable data on deaths and victims. Chen et al. (2020) contributed further by compiling educational data across provinces, highlighting discrepancies in educational spending that might contribute to long-term regional inequality. Other scholars have examined how political upheaval in general can disrupt economic progress. Acemoglu & Robinson (2013) noted that nations experiencing sudden political shifts often see long-term economic stagnation due to the destruction of institutional frameworks. Their work aligns with the patterns observed in this study, as the Cultural Revolution severely undermined regional governance structures. While these foundational studies provide insight into the socioeconomic consequences of political disruption, this research bridges a crucial gap by quantitatively linking the Cultural Revolution to long-term economic development. It provides fresh empirical evidence of the persistent economic cost of revolutionary policies and

calls for a renewed focus on safeguarding economic structures in politically volatile regions.

This research employs an instrumental variable approach, using radio signal reception as a proxy for revolutionary propaganda intensity. This approach is inspired by Ou & Xiong (2021), analyzing the impact of radio signal reception on revolutionary fervor, which motivates me to extend their framework by focusing on economic outcomes. This instrumental variable method is critical due to the endogenous relationship between revolution intensity and economic development. In this context, regions with better radio reception experienced more intense revolutionary activities, allowing radio signals to serve as an instrumental variable (IV) for revolution intensity. The data employed in this study is sourced from key datasets, including those by Walder (2020), Chen et al. (2020), Liu (1971), and the National Bureau of Statistics of China. The core variables in this analysis include GDP output per capita and agricultural productivity between 2004 and 2022. By incorporating provincial-level control variables and refining the analytical model, this study seeks to offer a detailed understanding of how historical revolutionary activities influence long-term economic development.

The empirical framework involves two stages: first, confirming the relevance of radio signals as an instrument by testing their correlation with the endogenous variable, revolution intensity; and second, establishing the causal relationship between revolution intensity (instrumented through radio signals) and long-term economic development. A vector of control variables is introduced to account for educational and demographic differences among provinces pre-dating and during the Cultural Revolution, respectively.

The regression analysis demonstrates a statistically significant relationship between revolution intensity and long-term economic outcomes. Specifically, greater revolution intensity, as indicated by higher radio signal reception, correlates negatively with GDP per capita and agricultural productivity. Provinces that experienced more intense revolutionary fervor exhibit significantly lower growth rate in GDP output and agricultural productivity in subsequent decades. The results underscore the lasting impact of revolutionary fervor on regional economic

structures. In particular, GDP per capita and agricultural productivity remain suppressed, even after controlling for demographic and educational variables.

In the first-stage analysis, radio signals are a strong predictor of revolution intensity, with a significant positive relationship. In the second-stage regression analysis, with control variables being added, increased revolutionary fervor leads to notable declines in economic performance. These results affirm the theoretical premise that political instability can have enduring adverse effects on economic productivity. However, while the use of radio signals presents an innovative approach to addressing endogeneity concerns, the strength and reliability of this instrumental variable require further examination. As such, the concluding sections of this paper discuss the limitations of the current methodology and suggest directions for future research, including the need for additional control variables to enhance the robustness of the causal inference.

Overall, this study makes a significant contribution to the growing body of literature by exploring the long-term economic impact of socio-political movements. The findings confirm that revolutionary intensity during the Cultural Revolution had substantial and enduring negative effects on economic development in China. These results underline the importance of maintaining resilient institutional frameworks to minimize the long-term economic fallout from such political upheavals.

The rest of the paper is organized as follows: Section 2 provides background information on the Cultural Revolution, Section 3 provides a detailed literature review on the Cultural Revolution and economic performance, Section 4 discusses the empirical framework, Section 5 presents the datasets used in this study, Section 6 displays the findings, and Section 7 concludes the study.

2. The Cultural Revolution

The Cultural Revolution is one of the most pivotal and transformative events in modern

Chinese history, lasting from 1966 to 1976. It was launched by Mao Zedong, the Chairman of the Chinese Communist Party, during his last decade in power. The main goal of the Cultural Revolution is to reduce inequality among social strata and thus, provide equal opportunities to different people in terms of education, health care, and culture (Lieberthal, 2024; MacFarquhar & Schoenhals, 2008). During the Cultural Revolution, Chairman Mao initiated an unprecedented action by encouraging the Red Guards, a revolutionary organization made up of ordinary young civilians, to test the values of the officials from the communist party as well as intellectuals and scientists by criticizing them in public (Lieberthal, 2024). This resulted in violence, humiliation, torture, and, in some cases, the deaths of tens of thousands of people. According to the estimates, between 750,000 and 1.5 million people were killed during the Cultural Revolution, and 36 million people suffered from some form of political repression (Walder & Su, 2003). The movement was marked by extensive political, economic, and cultural chaos that left deep and enduring scars on Chinese society. The following subsections discuss the motivations, dynamics, and aftermath of the Cultural Revolution, which provides essential context for analyzing its long-term economic impacts.

2.1. The Objectives of the Revolution

Chairman Mao Zedong envisioned the Cultural Revolution as a means to solidify the Maoist ideology, especially facing perceived threats from emerging factions within the Chinese Communist Party and rising economic disparities. The Revolution was officially launched in 1966 after the publication of a directive known as the "May 16 Notice", which accused internal elements of attempting to subvert socialism. Mao called upon the masses, particularly the youth, to defend the revolution by purging the "Four Olds": old customs, old culture, old habits, and old ideas.

2.2. The Red Guards and Political Repression

Central to the revolution's implementation was the mobilization of millions of students and young workers into the Red Guards, a paramilitary group tasked with identifying and eliminating counter-revolutionaries. The Red Guards, fervently devoted to Mao, turned against intellectuals, educators, artists, and even government officials by accusing them of supporting counter-revolutionary beliefs. Their activities were fueled by intense propaganda, leading to widespread social instability and mass repression. Millions of lives were affected through humiliation, torture, and execution. Official estimates indicate that as many as 1.5 million people were killed, and up to 36 million suffered political persecution (Walder & Su, 2003).

2.3. Cultural Devastation and the "Four Olds" Campaign

The effect of the Cultural Revolution on China's cultural heritage was systemic and devastating. The "Four Olds" campaign aimed to dismantle traditional symbols that contradicted the revolutionary ideology. Temples, churches, historical sites, and classical art were destroyed. Books and other literature representing traditional values or bourgeois sentiment were burned. This campaign effectively erased much of China's rich cultural heritage and historical artifacts.

2.4. The "Down to the Countryside" Movement and Rural Dislocation

In 1968, the Cultural Revolution entered a new phase when Mao launched the "Down to the Countryside" movement, forcing over 16 million urban youth to relocate to rural areas. The official justification for this movement was to expose urban youth to the realities of rural life and teach them socialist values by "learning from the peasants". However, it resulted in significant social disruption as the urban youth struggled to integrate into rural communities. They faced harsh living conditions, unfamiliar agricultural work, and profound disillusionment. This movement also resulted in educational and occupational displacement, limiting future opportunities for an entire generation.

2.5. Economic Consequences and Institutional Damage

The Cultural Revolution's economic consequences were significant and far-reaching. Industrial production, scientific research, and education came to a standstill as the nation focused on political campaigns rather than economic development. The purging of intellectuals and party officials disrupted institutional continuity, leading to inefficiency and a vacuum of expertise in key sectors. Production quotas fell, agricultural productivity lagged, and regional disparities in economic development widened.

At the institutional level, the Revolution weakened the central authority of government and dismantled organizational structures which are crucial for economic planning. Local leaders often acted autonomously to safeguard their interests, causing economic policies to be inconsistently implemented across provinces.

2.6. Aftermath and Reforms

The Cultural Revolution formally ended with Mao Zedong's death in 1976 and the arrest of the Gang of Four, a radical political faction responsible for executing many of the Revolution's excesses. In the following years, Deng Xiaoping initiated sweeping reforms that emphasized pragmatic economic policies over ideological purity. These reforms gradually led to China's dramatic economic transformation. Despite China's remarkable economic growth since the late 1970s, the Cultural Revolution's long-term impact is still felt. It shaped generational attitudes toward government and authority, left enduring issues on education, and created economic disparities across regions.

In summary, the Cultural Revolution was not only a socio-political upheaval but also a period of profound economic dislocation. It continues to influence contemporary Chinese society and development. Understanding the Revolution's regional impacts and its influence on economic indicators is crucial for comprehending China's historical development. This study

aims to explore the relationship between revolutionary intensity and long-term economic outcomes across Chinese provinces.

3. Existing Literature on the Cultural Revolution and Economic Development

The Cultural Revolution, characterized by widespread political upheaval, has been extensively studied for its profound impacts on society and the economy. The existing literature, while rich, often focuses on specific aspects of the Revolution such as mass persuasion, political repression, Red Guard activities, educational disruptions, and the implementation of significant policies and campaigns, each contributing uniquely to modern China's economic outcomes. This literature review seeks to combine these studies and compare them with the novel contributions of this study which aims to explore a wider range of economic outcomes and offer provincial insights into the economic impacts of the Cultural Revolution.

Mass persuasion during the Cultural Revolution, as investigated by Ou & Xiong (2021), shows that regions with stronger radio signals had higher exposure to propaganda, and thus experienced more intense revolutionary activities. This aligns with Strömberg's (2004) findings in a different context, where U.S. counties with higher radio listenership witnessed greater public spending from the government during the Great Depression. These pieces of evidence suggest that media plays a broad and influential role in initiating political and economic responses.

Political repression has been another focus of previous research. Studies by Wang (2019) and Wang (2017) highlight the long-term effects of the Cultural Revolution's violence on political attitudes and behaviors of individuals, which indicates a sustained distrust in political leaders and skepticism towards politics. This finding is paralleled in studies conducted by Pop-Eleches & Way (2021) and deMeritt (2016), which discuss how political repression can help reach the

government's objectives strategically or affect group behaviors based on the nature of rewards, further explored by Bueno De Mesquita & Shadmehr (2022).

Red Guard activities have been widely documented, representing an essential aspect of the Cultural Revolution. Fueled by Chairman Mao's instructions, these activities resulted in severe disruptions socially and culturally and contributed to an atmosphere of fear and compliance (Wasserstrom, 2017; Xu, 2010; Dongen, 2022; Israel, 1970).

However, these studies mentioned in previous paragraphs do not fully explore how mass persuasion, political repression, or Red Guard activities during the Cultural Revolution affected economic variables across different regions. To fill the research gap, this study intends to analyze how various degrees of political violence influenced economic development, adding a significant extension beyond the typical focus of existing literature.

Several journal articles have studied the economic drawbacks caused by the Cultural Revolution in terms of education and key policy implementations. The impact on education during the Revolution was notably detrimental. Research by Meng & Gregory (2002) and Meng & Zhao (2017) discusses how educational disruptions prevented academic progress and future income potential, which is consistent with earlier studies highlighting the long-term economic drawbacks of interrupted education (Zhou, 2016; Meng & Gregory, 2007).

Key policy implementations such as the "Four Olds" campaign and the "Down to the Countryside" Movement also had significant economic implications. Kraus (2011) and Kaczvinsky (2016) explore how these campaigns not only altered cultural landscapes but also influenced economic behaviors and expectations. Chen et al. (2020) provides a more positive view of the "Down to the Countryside" Movement, suggesting that interactions between urban youth and rural populations during these movements enhanced educational outcomes and economic opportunities in rural China.

Nevertheless, while educational disruption has been linked to lower future income potential in prior research (Meng & Gregory, 2002; Zhou, 2016), this study seeks to expand on

this topic by exploring other economic factors influenced by educational setbacks, such as regional agricultural capabilities. This allows for a more comprehensive understanding of the economic performance of educational disruptions during the Cultural Revolution.

In addition to previous studies which mainly focus on the Revolution, Bai & Wu (2023) present a broader economic analysis by controlling for pre-revolution outcomes and using transitory shocks to agricultural income during the Revolution as an instrumental variable, arguing that counties with higher revolutionary intensity experienced lower industrialization levels and economic growth. This underscores a complex relationship between revolution and economic development, highlighting the disruptive impact of intense revolutionary activities on economic stability and growth.

In contrast to these previous works, this study seeks to fill several gaps in the existing literature by employing a more expansive set of economic indicators to assess the impacts of the Cultural Revolution. By examining not only GDP per capita but also agricultural productivity, this study aims to provide a more comprehensive analysis of the Revolution's economic aftermath. Furthermore, this research introduces an innovative methodology by using radio signals as an instrumental variable to examine causal relationships between the revolution intensity and economic outcomes. This not only differentiates it from prior studies which may lack such methodological rigor but also enhances the potential to uncover insights into how various degrees of revolutionary propaganda influenced economic development across different provinces. By focusing on regional variations, this study also adds a new dimension to our understanding of the Cultural Revolution's economic consequences by revealing patterns of economic resilience or decline that are masked by national-level data, thus providing valuable insights for regional policy-making and economic strategy.

In sum, this literature review demonstrates that while existing studies provide a valuable foundation for understanding the impacts of the Cultural Revolution on various facets of Chinese society, significant gaps remain particularly in linking political and social upheavals to a

broader economic context. This study addresses these gaps by offering more detailed economic analysis and introducing a new methodological approach, thereby offering a fresh perspective on the regional variations in the economic aftermath of the Cultural Revolution and potentially guiding future policy decisions aimed at fostering economic resilience and growth in regions most affected by historical upheavals.

4. Empirical Framework

The empirical framework for analyzing the causal relationship between the intensity of the Cultural Revolution and long-term economic development in China is built upon the use of instrumental variables (IV) regression analysis. This approach helps to address potential endogeneity issues by isolating the exogenous variations that are only correlated with the independent variable of interest (revolution intensity) and not with the error term in the regression model.

4.1. Ideal Datasets

The ideal datasets should include comprehensive metrics on revolution intensity such as the number of deaths and the number of victims. Additionally, economic development indicators like GDP per capita and agricultural productivity for each province would be crucial. However, challenges might arise from data inaccuracies due to underreporting during the Cultural Revolution and disparities in how data was collected across regions. Furthermore, since the dataset spans a significant historical period, changes in data collection methods over time could introduce inconsistencies.

4.2. Theoretical Concepts and Measurement

Revolution intensity is measured using data from regional gazetteers that detail incidents

during the Cultural Revolution. It provides a direct measure of the sociopolitical upheaval during the period via the number of deaths and victims. Economic development is measured through long-term indicators such as GDP per capita and agricultural productivity. These indicators are measured several decades after the Revolution to capture the enduring impacts.

4.3. Econometric Techniques (IV)

The number of radio stations in each province before the Cultural Revolution (1962) is used as an instrumental variable for revolution intensity. This choice is inspired by the work of Ou & Xiong (2021), who demonstrated that radio signals could intensify revolutionary activities due to increased propaganda dissemination.

Regression Model Specifications

First Stage Regression:

$$(Revolution\ Intensity)_i = \gamma_0 + \gamma_1 (Radio\ Signal)_i + \gamma_2\, Z_i + \varepsilon_i$$

where $(Radio\ Signal)_i$ is the instrumental variable used to predict the endogenous variable $(Revolution\ Intensity)_i$, Z_i is a vector of control variables that capture other factors potentially influencing revolution intensity, and ε_i is the error term.

Second Stage Regression:

$$Y_i = \beta_0 + \beta_1 (Revolution\ Intensity)_i + \beta_2\, X_i + \epsilon_i$$

where the predicted values of $(Revolution\ Intensity)_i$ from the first stage are used to estimate the effect on the economic outcome Y_i, X_i is a vector of control variables that capture other factors potentially influencing economic development, and ϵ_i is the error term.

For the control variables, Z_i in the first stage could include historical factors such as population density or levels of education prior to the Revolution that might impact the intensity of revolutionary activities. X_i in the second stage could include contemporary factors relevant to

economic development, such as educational attainment, infrastructure, and demographic trends.

4.4. Justification for the Instrumental Variable

The relevance condition of the instrumental variable is tested by confirming the strong correlation between radio signals and revolution intensity. A strong first-stage relationship is crucial for the IV to provide a valid prediction. Also, the IV must be correlated with the outcome only through its association with the endogenous predictor (revolution intensity). Historical evidence and previous studies suggesting that radio was primarily a tool for spreading propaganda support the assumption that radio signals are exogenous to other factors influencing economic outcomes, such as pre-existing economic conditions. In addition, including controls enhances the model's accuracy in isolating the effect of revolution intensity on economic development by accounting for other factors that could influence both.

In sum, this empirical framework allows for a robust analysis of causal effects, mitigating biases that could arise from omitted variables. By systematically applying these econometric techniques, this study aims to provide deeper insights into the long-term economic impacts of the Cultural Revolution, contributing to the understanding of historical events and their legacy.

5. Data

This section discusses the datasets used in the analysis. TABLE 1 provides an overview of the description and method of calculating the main variables. There are 4 categories of variables used in this study – revolution intensity (independent variables), economic outcomes (dependent variables), a series of educational and demographic data (controls), and radio signals (instrumental variable). Detailed discussion is provided in the following subsections.

5.1. Revolution Intensity

The data on revolution intensity is derived from a comprehensive cross-sectional dataset compiled by Walder (2020), which examines the Cultural Revolution's intensity at the county level in 1967. I aggregate the county-level dataset into province-level for the purpose of this study. This dataset is based on regional gazetteers [3]that meticulously document various violent cases and their consequences throughout China during this chaotic period. Gazetteers are detailed, book-length reference works covering a region's local history, economy, and demographics. Following a central policy in 1978 that mandated county-level investigations into the Cultural Revolution, these gazetteers were published to rehabilitate victims and provide accurate historical accounts (Ou & Xiong, 2021; Su, 2011).

The following are the key variables used in this study. *Deaths (reported)* represents the total number of deaths directly linked to the Cultural Revolution, recorded through various sources and regional gazetteers. *Deaths (published)* represents the number of deaths reported in published documents and books specifically covering the Cultural Revolution. *Victims (all sources)* is broader measure including all individuals who were either directly harmed or perished during the Cultural Revolution, as compiled from a variety of sources. *Victims (published)* refers to the number of victims documented in published sources covering the period.

TABLE 2 provides a detailed provincial breakdown of deaths and victims, with provinces like Guangdong, Guangxi, and Yunnan recording the highest numbers of victims. This indicates significant regional variation in the revolution's impact. FIGURE 1 illustrates the intensity of the Cultural Revolution varied across counties, with darker shades representing higher death rates and thus greater revolution intensity. This geographic variation demonstrates how the violence and upheaval were not uniform but were concentrated in particular regions.

[3] See Walder (2014) which discusses how the extent of violence is documented, using county-level gazetteers in detail

Walder's dataset is one of the most comprehensive collections available that quantifies the Cultural Revolution's impact, as it meticulously aggregates data at county levels. This dataset also serves as a foundation for understanding the socio-economic and demographic ramifications of the Cultural Revolution, helping researchers analyze the legacy of this significant period in Chinese history. By offering detailed provincial insights, this dataset supports studies on the regional patterns and consequences of the Cultural Revolution and provides a robust framework for evaluating its long-term impacts on Chinese society.

5.2. Economic Outcomes

The economic outcomes following the Cultural Revolution is examined using provincial GDP per capita and agricultural productivity data from 2004 to 2022 from the National Bureau of Statistics of China. The analysis aims to investigate how the Cultural Revolution may have influenced long-term economic trajectories across different provinces in China.

TABLE 3 presents GDP per capita at the province level, with units in 0.1 million yuan per person. It includes summary statistics such as mean, standard deviation (sd), minimum (min), and maximum (max) values for each province. Provinces like Shanghai and Beijing demonstrate the highest averages, while Gansu and Guizhou have the lowest mean GDP per capita. TABLE 4 presents agricultural productivity in terms of the gross output value of agriculture, forestry, animal husbandry, and fishery across provinces, measured in 100 million yuan. The summary statistics offer a snapshot of the variation in agricultural output, with provinces like Shandong, Henan, and Sichuan consistently showing high mean output levels. The standard deviation highlights significant variability within provinces, indicating regions where agricultural productivity may have fluctuated widely. TABLE 3 and TABLE 4 illustrate disparities in economic performance across different regions. Coastal provinces generally show higher GDP per capita and agricultural productivity, while inland provinces often lag behind. The high variation in agricultural productivity suggests that provinces have unique challenges and advantages that

affect their economic development.

These metrics on economic outcomes provide crucial insights into how regions with varying degrees of Cultural Revolution intensity developed in the long run. By comparing these data points with the intensity of the Cultural Revolution (as measured in Section 5.1.), researchers can assess whether provinces more severely affected by the revolution experienced different economic trajectories. The datasets highlight the importance of tailored economic policies that consider each province's unique economic conditions and history. Besides, this comprehensive dataset allows for an in-depth study of regional economic trends in China, helping identify patterns that may have persisted from the period following the Cultural Revolution up to the present day.

5.3. Control Variables

To ensure a comprehensive understanding of the variation among provinces before and after the Cultural Revolution, this study employs a province-level educational dataset ranging from 1950 to 1965 to establish a baseline for pre-revolution comparisons (TABLE 5). The educational dataset, originally sourced from local gazetteers and compiled by Chen et al. (2020), is considered the most relevant data related to economic development available. In addition, this study employs a province-level demographic dataset in 1967 to control demographic factors during the period of the Cultural Revolution (TABLE 6). This dataset is originally sourced from local gazetteers and compiled by Walder (2020) and is considered the most relevant data related to revolution intensity available.

5.4. Radio Signal

The dataset from Liu's (1971) work – Communications and National Integration in Communist China – provides crucial information on the distribution of radio stations across Chinese provinces in 1962, just prior to the onset of the Cultural Revolution in 1966. This

provincial-level cross-sectional data shows the prevalence and reach of radio signals before the Revolution's significant historical events. Additionally, this dataset allows researchers to analyze how radio signals, as an influential communication medium, may have impacted the intensity of the Cultural Revolution. A study by Ou & Xiong (2021) established that provinces with more radio stations experienced more intense revolutionary activities, indicating a causal relationship between the availability of radio signals and the Revolution's dynamics.

In terms of instrumental variables, this dataset's information on radio station distribution can serve as an instrumental variable in analyzing the Cultural Revolution's effects on subsequent economic development. The assumption is that provinces with more radio stations had their revolutionary activities influenced by heightened propaganda and mass communication.

The data in TABLE 7 offers a snapshot of radio station distribution across China's provinces in 1962, where Jiangsu and Hebei had the highest number of radio stations (11 each), followed by Inner Mongolia and Liaoning (10 each), Heilongjiang (9), and Jilin and Gansu (8 each). While certain provinces in the north and northeast had the most significant number of radio stations, other areas had fewer stations. Provinces like Xinjiang (2 stations), Yunnan (2), and Tibet (1) had limited access to radio communication. This distribution highlights the disparities in communication infrastructure across the nation, which may have shaped the extent to which different regions received revolutionary messaging.

Overall, the Liu (1971) dataset is pivotal in understanding the relationship between media infrastructure and socio-political events. By analyzing the distribution of radio stations in 1962, it becomes possible to uncover how communication affected the revolutionary fervor and its economic aftermath.

6. Results

This section presents the findings of the study via regression analysis. TABLE 8 examines

the correlation between the instrumental variables and the control variables. TABLE 9 and TABLE 10 present the results of the first stage and second stage regression analysis, respectively. Detailed discussion is provided in the following subsections.

6.1. Correlation Between Radio Signals and Controls

TABLE 8 presents the results from regressions analyzing correlations between the number of radio stations (in 1962) and several pre-revolution educational and demographic variables during the Revolution to determine whether radio signals are correlated with these controls, and thus, whether radio signals can be used as a valid instrumental variable. The following is a detailed interpretation of the findings.

In terms of significant findings, a statistically significant positive relationship is observed between the number of cadres and the number of radio stations (coefficient = 8.36, $p < 0.05$). This suggests that provinces with more radio stations tend to have a higher number of cadres, potentially reflecting the role of local authorities in establishing radio stations. In addition, the coefficient of 11.36 ($p < 0.10$) indicates a statistically significant positive correlation between the number of radio stations and the number of state-funded secondary school teachers. This relationship might suggest that regions with more teachers in such schools had better access to radio signals or placed greater emphasis on education.

For non-significant findings, the coefficients for the rest of the variables – urban population, number of workers, the population in 1967, the population in 1964, the number of primary schools, the number of secondary schools, fiscal education expenses, the number of teachers in primary schools, the number of teachers in state-funded primary schools, the number of teachers in non-state-funded primary schools, the number of teachers in secondary schools, and the number of teachers in non-state-funded secondary schools are all statistically insignificant. This implies no strong direct correlation between these variables and the prevalence of radio signals.

For the model fit, the overall model exhibits a high R^2 of 0.93, meaning that the combination of the control variables explains 93% of the variance in the number of radio stations. The F-statistic of 3.60 (Prob > F = 0.11) indicates that the model as a whole is not statistically significant at conventional levels.

Considering the majority of insignificant coefficients and low F-statistic, the results suggest that radio signals (measured via the number of radio stations) have relatively weak correlations with most control variables, aside from cadres and state-funded secondary school teachers. While these two significant correlations should be investigated further to understand any potential bias, the absence of strong correlations with other controls provides some confidence that the instrument can satisfy the exclusion restriction.

Overall, the results in TABLE 8 imply that radio signals have a generally random correlation with the majority of control variables and thus could serve as an appropriate instrumental variable for analyzing the impact of the Cultural Revolution's intensity on economic outcomes. Nevertheless, further investigation into the correlations with cadres and state-funded secondary teachers is needed to fully confirm instrument validity. Also, the small sample size resulting from the province-level analysis with cross-sectional datasets might generate inaccurate results.

6.2. First Stage Regression Analysis

The first-stage regression results presented in TABLE 9 are designed to examine how radio signals (approximated by the number of radio stations in 1962) relate to different measures of revolution intensity (deaths and victims, both reported and published) during the Cultural Revolution. These results are crucial for establishing the validity of radio signals as an instrumental variable for assessing the impact of revolution intensity on economic outcomes. The following is a detailed breakdown and interpretation of the findings.

In terms of the relationship between radio signals and revolution intensity, according to

Columns (1) and (3), for both reported and published deaths, there is a statistically significant positive relationship with the number of radio stations, where a one-unit increase in radio stations is associated with an 18% increase in both reported and published deaths. This suggests that regions with more radio stations, and presumably greater exposure to revolutionary propaganda, experienced more intense revolution. According to Columns (5) and (7), similar to reported and published deaths, there is a positive association for both uncontrolled scenarios of reported and published victims, with coefficients of 0.16 and 0.17 respectively. This indicates that increased radio presence correlates with higher numbers of victims. The statistical significance at the 10% level reinforces the role of radio as a potential indicator of revolutionary fervor.

Taking the impact of controls into account, Columns (2), (4), (6), and (8) show that when demographic and pre-revolution educational data are included as control variables, the coefficients associated with radio signals on all measures of revolution intensity become statistically insignificant and generally negative (e.g., -0.56 for the number of reported deaths with controls). This shift suggests that other confounding factors, possibly related to socioeconomic conditions or pre-revolution educational infrastructure, might explain the variation in revolution intensity more effectively than the number of radio stations alone.

For statistical significance and model fit, the R^2 values increase significantly when controls are included, jumping from values around 0.13 to 0.16 in the uncontrolled models to 0.59 to 0.93 in the controlled models. This improvement indicates that the controls are important in explaining the variation in revolution intensity across provinces. The large increase in R^2 values suggests a much better model fit when these variables are considered.

In terms of the instrument validity, the significant changes in coefficient signs and significance levels upon the introduction of controls raise concerns about the exclusivity of radio signals as an instrumental variable. A valid instrument should not be correlated with other factors that influence the dependent variable (revolution intensity) besides through the channel

of the instrument (radio signals). The negative coefficients in the controlled models suggest potential endogeneity issues where radio signals might be correlated with unobserved factors that also affect revolution intensity.

Additionally, there are several concerns in the analysis. (1) Small sample size: As the analysis is conducted on the province level with cross-sectional datasets, this leads to a small sample size, which might generate inaccurate results. (2) Endogeneity concern: The changes in sign and significance suggest that radio may not fully satisfy the exclusion restriction criteria, which states that the instrument must not be correlated with the error term in the main regression equation apart from its effect through the independent variable (revolution intensity). (3) Potential overfitting: The extremely high R^2 values in some controlled models might suggest overfitting, especially in historical data analyses where many unobserved factors could influence outcomes.

Overall, the initial findings suggest a potential instrumental role for radio signals in explaining revolution intensity due to the significant associations in the uncontrolled models. However, the role of radio signals as a valid instrument needs careful reevaluation considering the impact of controls, which significantly alter the results. This analysis should guide further methodological refinement and perhaps encourage the search for additional or alternative instruments to capture the causal impacts of the Cultural Revolution's intensity on economic outcomes in a more robust way.

6.3. Second Stage Regression Analysis

TABLE 10 presents the second stage regression results using radio signals as an instrumental variable to examine the impact of revolution intensity, as measured by deaths and victim counts, on economic outcomes (GDP per capita and agricultural output) with control variables included.

The coefficients for both the GDP and agricultural productivity models are consistently

negative across different measures of revolution intensity. This indicates that higher revolution intensity (i.e., more deaths and victims) is associated with lower economic outcomes in both GDP per capita and agricultural productivity. Moreover, deaths and victims both show significant impacts at various levels of significance. A one percent increase in reported deaths leads to a 11 percent decrease in GDP growth and a 16 percent decrease in the growth rate of agricultural output. Published deaths and all sources of victims follow a similar pattern, indicating robustness across different measures of revolution intensity.

The R^2 values suggest a good fit for the models, especially for GDP growth rate (ranging from 0.60 to 0.70), indicating that the models explain a significant proportion of the variability in economic outcomes. However, the relatively low F-statistics suggest that while radio signals serve as an instrumental variable, their strength might not be robust across all specifications, calling for cautious interpretation of the IV's strength as an identifier of causal effects. Also, the small sample size resulting from the province-level analysis with cross-sectional datasets might generate inaccurate results.

Overall, the results underscore the potentially detrimental long-term effects of the Cultural Revolution's intensity on China's economic development. The negative associations found in both the growth rate of GDP per capita and agricultural productivity highlight the social and political upheaval experienced during the Cultural Revolution in terms of the economy. These findings contribute valuable insights into the broader implications of political chaos on economic performance and development trajectories.

7. Conclusion

This study investigates the long-term economic impacts of the Cultural Revolution in China, focusing on the changes in GDP output and agricultural productivity from 2004 to 2022.

Using a novel empirical strategy, the study utilizes radio signals as an instrumental variable to assess the causal effects of the intensity of the Cultural Revolution on economic outcomes. The findings reveal that regions with more intense revolutionary activities, as indicated by higher deaths and victim counts, generally experienced poorer economic performance in the ensuing decades. These results underscore the profound and enduring economic consequence of the Cultural Revolution, highlighting its role in shaping China's modern economic outcomes.

While this study offers significant insights, it comes with limitations that suggest caution in interpreting the results. First, the reliability of radio signals as a robust instrumental variable is questionable. The variable's strength and validity as an instrument were inconsistent across different model specifications, indicating potential weaknesses in isolating the causal impact of revolution intensity. Additionally, this study faces challenges related to data quality and availability, typical of historical economic analyses, where inconsistencies in data collection methods over time could have introduced biases. Furthermore, the analysis was constrained by the scope of available data and limited sample size, primarily focusing on aggregate provincial outcomes, which may mask the significant variations among provinces.

Future research should aim to address these limitations by exploring additional instrumental variables that could more reliably isolate the causal impacts of cultural-political upheavals on economic outcomes. Researchers might consider using alternative data sources or methodologies to assess the impact of the Cultural Revolution, such as micro-level data that can capture individual or household impacts more precisely. Future studies could also involve a comparative analysis with other countries that have experienced similar large-scale socio-political disruptions to understand the broader implications of such events on economic development. In addition, future studies could expand on the range of economic indicators analyzed to include measures like income inequality, employment rates, and sector-specific productivity changes. This would provide a more comprehensive understanding of the economic influence of the Cultural Revolution. It would also be beneficial to integrate qualitative assessments that could offer

deeper insights into the mechanisms through which revolutionary activities influenced economic trajectories.

Exploring these ideas will not only deepen our understanding of the Cultural Revolution's impact but also enhance our broader comprehension of how major political and social upheavals can shape economic development, informing policy strategies aimed at fostering resilience and sustainable growth in the face of such challenges. This comprehensive approach will further enrich the economic literature, bridging historical events and modern economic analysis, and ultimately contribute to more informed policy-making and economic strategy development.

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Tables and Figures

TABLE 1: DESCRIPTION AND METHOD OF CALCULATING THE MAIN VARIABLES FOR ANALYSIS

Variable Type	Variable Name	Description	Calculation Method
Independent Variables	Death (reported)	The number of reported deaths in 1967	Take the logarithm
	Death (published)	The number of published deaths in 1967	Take the logarithm
	Victims (all sources)	The number of victims collected from all sources in 1967	Take the logarithm
	Victims (published)	The number of published victims in 1967	Take the logarithm
Dependent Variables	GDP	GDP per capita growth rate (2004 – 2022)	Compute the average GDP per capita from 2004 to 2022 and then use the logarithm
	Agriculture	Agricultural growth rate (2004 – 2022)	Compute the average agricultural output from 2004 to 2022 and then use the logarithm
Controls	Urban population	The number of populations living in urban area in 1967	Take the logarithm
	# of workers	The number of workers in 1967	Take the logarithm
	# of cadres	The number of cadres in 1967	Take the logarithm
	Population in 1967	Population in 1967	Take the logarithm
	Population in 1964	Population in 1964 (pre-revolution)	Take the logarithm
	# of primary schools	The number of primary schools (pre-revolution)	Take the logarithm
	# of secondary schools	The number of secondary schools (pre-revolution)	Take the logarithm
	Fiscal education expenses	Fiscal education expenses (pre-revolution)	Take the logarithm
	# of teachers in primary schools	The number of teachers in primary schools (pre-revolution)	Take the logarithm
	# of teachers in state-funded primary	The number of teachers in state-funded primary schools (pre-revolution)	Take the logarithm
	# of teachers in non-state-funded primary	The number of teachers in non-state-funded primary schools (pre-revolution)	Take the logarithm
	# of teachers in secondary schools	Number of teachers in secondary schools (pre-revolution)	Take the logarithm
	# of teachers in state-funded secondary	Number of teachers in state-funded secondary schools (pre-revolution)	Take the logarithm
	# of teachers in non-state-funded secondary	Number of teachers in non-state-funded secondary schools (pre-revolution)	Take the logarithm
IV	Radio	The number of radio stations	

TABLE 2: SUMMARY STATISTICS OF INDICATORS ON REVOLUTION INTENSITY (PROVINCE-LEVEL), 1967

Province	Deaths (reported)	Deaths (published)	Victims (all sources)	Victims (published)
Anhui	638	638	135,670	135,670
Beijing	5,807	5,807	336,279	336,279
Fujian	1,713	1,713	185,632	185,632
Guangdong	31,274	31,274	1,001,518	1,001,518
Gansu	3,714	3,714	360,728	360,728
Guangxi	82,868	55,651	745,475	507,056
Guizhou	1,322	1,322	162,568	162,568
Hebei	16,734	16,734	989,144	980,628
Heilongjiang	3,060	3,060	229,655	229,655
Henan	1,378	1,378	370,718	370,718
Hubei	1,776	1,776	170,624	131,202
Hunan	14,165	7,889	627,458	594,581
Jilin	6,703	6,703	194,914	194,914
Jiangsu	3,877	3,877	1,322,026	1,190,452
Jiangxi	4,323	4,323	279,396	279,296
Liaoning	20,554	20,554	337,542	337,542
Inner Mongolia	10,646	10,646	475,125	475,125
Ningxia	388	388	25,914	25,914
Qinghai	254	254	17,494	17,494
Shaanxi	10,680	10,680	918,142	900,275
Sichuan	10,845	10,845	641,781	641,781
Shandong	2,033	2,033	482,013	482,013
Shanghai	11,588	11,588	721,801	721,801
Shanxi	3,753	3,753	191,095	188,222
Tibet	145	145	9,324	9,324
Xinjiang	3,212	3,212	199,952	199,952
Yunnan	14,763	14,077	1,501,998	1,443,230
Zhejiang	7,181	5,696	784,040	660,462

TABLE 3: SUMMARY STATISTICS OF GDP PER CAPITA (PROVINCE-LEVEL), 2004-2022

Province	Mean	Sd	Min	Max
Gansu	2.28	1.13	0.65	4.46
Guizhou	2.46	1.58	0.42	5.19
Guangxi	2.69	1.4	0.68	5.19
Tibet	2.87	1.64	0.79	5.91
Yunnan	2.91	1.71	0.71	6.08
Heilongjiang	2.95	1.19	1.08	5.11
Qinghai	3.1	1.64	0.82	6.09
Hebei	3.25	1.35	1.11	5.66
Jilin	3.25	1.5	0.91	5.54
Jiangxi	3.35	1.89	0.79	6.89
Sichuan	3.36	1.89	0.78	6.76
Henan	3.37	1.64	0.87	5.9
Shanxi	3.4	1.68	1.05	7.35
Hainan	3.44	1.77	0.98	6.71
Ningxia	3.44	1.78	0.88	7.01
Xinjiang	3.53	1.71	1.11	6.97
Anhui	3.58	2.07	0.82	7.28
Hunan	3.69	2.01	0.83	7.2
Shaanxi	4.07	2.24	0.85	8.3
Liaoning	4.12	1.57	1.53	6.87
Hubei	4.52	2.57	0.97	9.02
Chongqing	4.53	2.54	1.1	8.89
Inner Mongolia	4.72	2.41	1.23	9.74
Shandong	4.79	2.13	1.45	8.62
Guangdong	5.82	2.47	2.05	10.23
Fujian	6.18	3.38	1.62	12.36
Zhejiang	6.62	2.86	2.33	11.87
Tianjin	6.92	2.81	2.56	11.84
Jiangsu	7.52	3.8	1.97	14.34
Shanghai	10.38	4.21	4.42	18.1
Beijing	10.65	4.61	4.19	19.02

Notes: Units in 0.1 million yuan per person

TABLE 4: SUMMARY STATISTICS OF AGRICULTURAL OUTPUT (PROVINCE-LEVEL), 2004-2022

Province	Mean	Sd	Min	Max
Tibet	143.95	64.65	62.74	278.62
Shanghai	290.19	32.93	233.39	343.78
Qinghai	290.97	149.53	86.65	566.21
Beijing	315.01	54.14	240.19	421.78
Tianjin	347.11	89.31	225.04	521.44
Ningxia	419.1	209.16	125.52	845.92
Hainan	1162.71	535.99	438.73	2272.04
Shanxi	1238.11	524.4	441.85	2211.59
Gansu	1292.77	624.18	477.35	2680.74
Chongqing	1538.01	778.5	575.24	3068.45
Jilin	2020.62	633.98	940.67	3217.91
Guizhou	2223.72	1503.06	524.64	4908.67
Inner Mongolia	2389.28	966.63	851.3	4316.76
Shaanxi	2411.98	1207.81	651.21	4601.93
Jiangxi	2495.89	961.96	1054.96	4223.81
Zhejiang	2554.96	771.91	1332.27	3752.31
Xinjiang	2616.56	1408.51	750.68	5469.04
Fujian	3121.85	1316.57	1317.28	5502.56
Yunnan	3168.91	1749.83	965.22	6635.8
Liaoning	3396.48	1091.19	1510.52	5180.03
Anhui	3701.18	1455.42	1644.42	6277.98
Guangxi	3770.52	1663.74	1294.53	6938.53
Heilongjiang	3965.16	1929.18	1136.64	6718.24
Hunan	4519.39	1853.51	1913.31	8160.13
Hubei	4756.45	2151.98	1695.44	8939.33
Hebei	4804.06	1507.4	2375.89	7667.41
Guangdong	4979.85	2005.57	2154.79	8892.29
Jiangsu	5629.4	2047.2	2417.63	8733.8
Sichuan	5675.84	2318.09	2252.28	9859.75
Henan	6589.73	2336.35	2963.92	10952.24
Shandong	7745.81	2515.96	3453.91	12130.71

Notes: The agricultural productivity here refers to the gross output value of agriculture, forestry, animal husbandry, and fishery, with units in 100 million yuan.

TABLE 5: SUMMARY STATISTICS OF THE PROVINCE-LEVEL EDUCATIONAL DATA, 1950-1965

Variables	Mean	Sd	Min	Max
# of primary schools	144,987.80	100,466.60	359	340,825
# of secondary schools	2,493.89	1,865.57	0	8,756
Fiscal education expenses	15,872.30	12,994.88	1.63	44,878.95
# of teachers in primary schools	138,166.60	103,961.20	492	380,053
# of teachers in state-funded primary	35,958.68	34,512.10	0	127,401
# of teachers in non-state-funded primary	8,881.54	7,994.49	0	27,053
# of teachers in secondary schools	15,067.64	10,801.58	83	44,912
# of teachers in state-funded secondary	3,338.36	3,700.79	0	14,869
# of teachers in non-state-funded secondary	425.11	902.42	0	4,551
Population in 1964	1.81×10^{8}	1.24×10^{8}	0	4.89×10^{8}

Notes: Number of Primary Schools refers to the count of primary schools in each province. *Number of Secondary Schools* refers to the count of secondary schools in each province. *Fiscal Education Expenses* refers to the total fiscal expenditures on education in each province. *Number of Teachers in Primary Schools* refers to the total number of primary school teachers. *Number of Teachers in State-Funded Primary Schools* refers to teachers employed in government-funded primary schools. *Number of Teachers in Non-State-Funded Primary Schools* refers to teachers working in privately funded primary schools. *Number of Teachers in Secondary Schools* refers to the total number of secondary school teachers. *Number of Teachers in State-Funded Secondary Schools* refers to teachers in government-funded secondary schools. *Number of Teachers in Non-State-Funded Secondary Schools* refers to teachers in privately funded secondary schools. The unit of fiscal education expenses is 100 million RMB.

TABLE 6: SUMMARY STATISTICS OF DEMOGRAPHIC DATA, 1967

Variables	Mean	Sd	Min	Max
Urban population	3,979,715	2,590,425	106,537	9,854,094
# of workers	1,244,079	911,212.80	10,808	3,377,968
# of cadres	202,264.40	123,316.80	6637	483,054
Population in 1967	2.75×10^{7}	1.81×10^{7}	1,053,930	7.36×10^{7}

Notes: Urban Population refers to the total number of people living in urban areas. *Number of Workers* refers to the total number of people engaged in various labor sectors. *Number of Cadres* refers to the total number of administrative or managerial government employees. *Population in 1967* refers to the total population across each province.

TABLE 7: DISTRIBUTION OF RADIO STATIONS, 1962

Province	Number of Stations
Jiangsu	11
Hebei	11
Inner Mongolia	10
Liaoning	10
Heilongjiang	9
Jilin	8
Gansu	8
Henan	7
Hunan	7
Anhui	5
Zhejiang	5
Jiangxi	5
Shanxi	5
Guangxi	5
Hubei	4
Fujian	4
Shaanxi	4
Shandong	4
Guangdong	4
Sichuan	4
Guizhou	3
Qinghai	3
Xinjiang	2
Yunnan	2
Tibet	1

Source: Liu (1971)

TABLE 8: CORRELATION BETWEEN RADIO SIGNALS AND CONTROL VARIABLES

	Radio
Urban population	-3.27
	(4.90)
Number of workers	-2.18
	(3.42)
Number of cadres	8.36**
	(2.97)
Population in 1967	0.34
	(3.40)
Population in 1964	0.43
	(4.46)
Number of primary schools	2.35
	(1.91)
Number of secondary schools	-3.82
	(3.72)
Fiscal education expenses	0.74
	(0.94)
Number of teachers in primary schools	2.06
	(4.06)
Number of teachers in state-funded primary	-8.45
	(4.46)
Number of teachers in non-state-funded primary	-3.99
	(1.86)
Number of teachers in secondary schools	-0.91
	(3.18)
Number of teachers in state-funded secondary	11.36*
	(4.34)
Number of teachers in non-state-funded secondary	-0.99
	(0.71)
N	19
F	3.60
Prob > F	0.11
R^2	0.93

Notes: Two categories of controls are included in the analysis – pre-revolution educational data and demographic data during the Revolution. The pre-revolution educational data is obtained from Chen et al. (2020). Although this is not ideal, this is the most relevant pre-revolution dataset to the economic performance I can access online without permission. This is a panel dataset ranging from 1950 to 1990 on the county level. I select the period from 1950 to 1965 (before the Revolution happened) and aggregate the panel dataset into a province-level cross-sectional dataset. The variables included as controls in the analysis are the population in 1964, the number of primary schools, the number of secondary schools, fiscal education expenses, the number of teachers in primary schools, the number of teachers in state-funded primary schools, the number of teachers in non-state-funded primary schools, the number of teachers in secondary schools, the number of teachers in state-funded secondary schools, and the number of teachers in non-state-funded secondary schools. The demographic data during the Revolution is obtained from Walder (2020). In this study, I aggregate the county-level data into province-level and generate a cross-sectional dataset in 1967. The variables included as controls in the analysis are the urban population, the number of workers, the number of cadres, and the total population. *Radio* refers to the number of radio stations in each province in 1962, which is obtained from a cross-sectional dataset in Liu (1971). The number of radio stations serves as the independent variable and the controls serve as the dependent variables in the analysis. The logarithm is taken for all the controls in the analysis. Standard errors are included in the parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

TABLE 9: FIRST STAGE REGRESSION RESULTS OF REVOLUTION INTENSITY ON RADIO SIGNALS

	Deaths (reported)		Deaths (published)		Victims (all sources)		Victims (published)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Radio	0.18*	-0.56	0.18*	-0.54	0.16*	-0.19	0.17*	-0.15
	(0.09)	(0.67)	(0.09)	(0.61)	(0.09)	(0.18)	(0.09)	(0.16)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
N	25	19	25	19	25	19	25	19
R^2	0.13	0.59	0.14	0.63	0.16	0.93	0.16	0.93

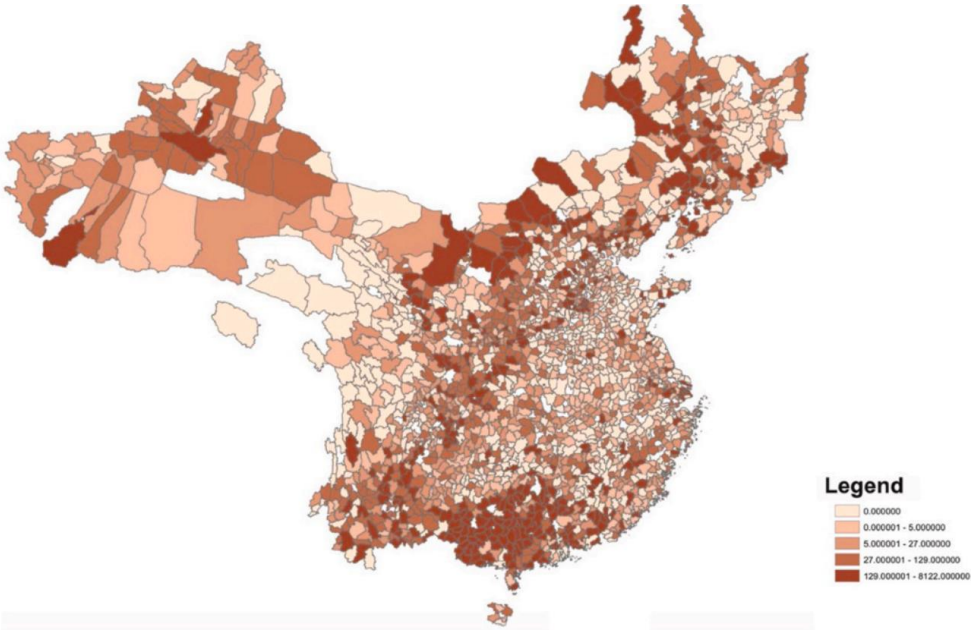
Notes: In the first row, *Deaths (reported)* represents the reported number of deaths during the Revolution, *Deaths (published)* represents the number of deaths during the Revolution collected from published resources, *Victims (reported)* represents the number of reported victims during the Revolution, and *Victims (published)* represents the number of victims during the Revolution collected from published resources. These data are obtained from Walder (2020) on the county level. In this study, I aggregate the county-level data into province-level and generate a cross-sectional dataset in 1967. *Radio* refers to the number of radio stations in each province in 1962, which is obtained from a cross-sectional dataset in Liu (1971). Controls include two categories – pre-revolution educational data and demographic data during the Revolution. The pre-revolution educational data is obtained from Chen et al. (2020). Although this is not ideal, this is the most relevant pre-revolution dataset to the economic performance I can access online without permission. This is a panel dataset ranging from 1950 to 1990 on the county level. I select the period from 1950 to 1965 (before the Revolution happened) and aggregate the panel dataset into a province-level cross-sectional dataset. The variables included as controls in the analysis are the population in 1964, the number of primary schools, the number of secondary schools, fiscal education expenses, the number of teachers in primary schools, the number of teachers in state-funded primary schools, the number of teachers in non-state-funded primary schools, the number of teachers in secondary schools, the number of teachers in state-funded secondary schools, and the number of teachers in non-state-funded secondary schools. The demographic data during the Revolution is obtained from Walder (2020). In this study, I aggregate the county-level data into province-level and generate a cross-sectional dataset in 1967. The variables included as controls in the analysis are the urban population, the number of workers, the number of cadres, and the total population. The logarithm is taken for all the controls in the analysis. The first row refers to the dependent variables in the regression analysis where each variable represents a single regression and the logarithm is taken in the analysis. The number of radio stations serves as the independent variable. Standard errors clustered at the province level are included in the parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

TABLE 10: SECOND STAGE REGRESSION RESULTS USING RADIO SIGNALS AS AN INSTRUMENT, WITH CONTROL VARIABLES

	GDP				Agriculture			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log deaths (reported)	-0.11*				-0.16**			
	(0.07)				(0.08)			
Log deaths (published)		-0.11*				-0.17**		
		(0.06)				(0.07)		
Log victims (all sources)			-0.32*				-0.48***	
			(0.17)				(0.17)	
Log victims (published)				-0.42*				-0.62**
				(0.25)				(0.26)
F	0.71	0.79	1.12	0.82	0.71	0.79	1.12	0.83
N	19	19	19	19	19	19	19	19
R^2	0.62	0.66	0.70	0.60	0.43	0.51	0.66	0.51

Notes: *GDP* refers to the logarithm of the average GDP per capita from 2004 to 2022. *Agriculture* refers to the logarithm of the average agricultural output from 2004 to 2022. The data on GDP per capita and agricultural productivity are obtained from the National Bureau of Statistics of China. *Deaths (reported)* represents the reported number of deaths during the Revolution, *Deaths (published)* represents the number of deaths during the Revolution collected from published resources, *Victims (reported)* represents the number of reported victims during the Revolution, and *Victims (published)* represents the number of victims during the Revolution collected from published resources. These data are obtained from Walder (2020) on the county level. In this study, I aggregate the county-level data into province-level and generate a cross-sectional dataset in 1967. The F-statistic is used to test if radio signals are a strong IV. The first row refers to the dependent variables in the regression analysis where each variable represents a single regression and the logarithm is taken in the analysis. Standard errors clustered at the province level are included in the parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

FIGURE 1: VARIATION IN REVOLUTION INTENSITY AT THE COUNTY LEVEL



Notes: This figure displays the differences in revolution intensity at the county level. The darker color represents a higher death rate and thus, a higher revolution intensity in that area.
Source: Ou & Xiong (2021)